Shimony took a position in MIT's philosophy department while still working on his physics PhD with Wigner and started teaching classes there on the foundations of quantum physics, aimed at upper-level undergraduate students. He also made friends in the physics and philosophy departments at several of the other universities in the Boston area, so he was not terribly surprised when, sometime in the 1964–1965 academic year, he received an envelope containing a preprint from Brandeis University written by a physicist visiting there from CERN named John Bell. "I thought, 'Here's another kooky paper that's come out of the blue.' I'd never heard of Bell," Shimony recalled. "And it was badly typed, and it was on the old multigraph paper, with the blue ink that smeared. There were some arithmetical errors. I said, 'What's going on here?' But I re-read it, and the more I read it, the more brilliant it seemed. And I realized, 'This is no kooky paper. This is something very great.'"

Shimony started thinking about how to test Bell's theorem in the laboratory "almost immediately," by his own recollection. "As soon as I'd understood what he had done, I thought 'Now, that's really interesting.… Have the predictions of quantum mechanics been examined carefully in such situations?' Then I thought I knew one other relevant piece of literature." Shimony asked his friend Aharonov whether Wu's old experiment could be modified to test Bell's theorem; Aharonov told him (incorrectly) that the experiment already provided such a test. "Aharonov is a very fast thinker and a very fast talker, and I was in awe of him," Shimony recalled. "I… thought, 'He's right. Maybe he's right. But maybe he isn't right.' The more I thought of it, the less convinced I was."

Shimony worked on understanding the issue on and off for several years, not getting anywhere for a while, until 1968. That year, he moved to Boston University for his dream job: a joint appointment in the Physics and Philosophy Departments. Shortly thereafter, he took on a physics graduate student named Michael Horne and assigned him the task of looking into how to test Bell's theorem. "The more he read, the less optimistic he was that Wu's experiment could be used to test Bell's Inequality," Shimony recalled. Shimony and Horne hit the library and soon found the Kocher-Commins experiment, which Shimony immediately recognized was what they needed. "By March 1969 the main lines of the work of Horne and myself were complete," Shimony said. "I told Mike Horne… nobody else is